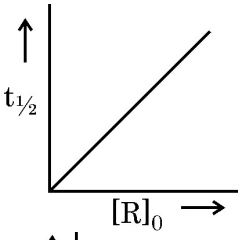
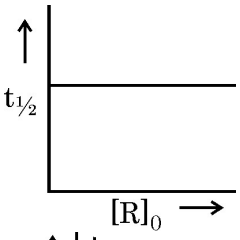
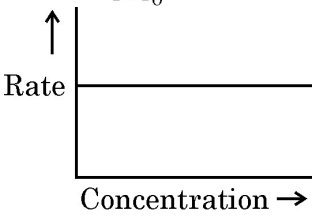
## SECTION – A

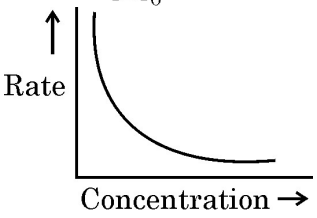
Questions No. **1** to **16** are Multiple Choice type questions carrying **1** mark each.

1. Which of the following curve represents the first order reaction ? **1**

(A) 


(B) 


(C) 


(D) 


2. Which of the following solutions will have the lowest freezing point in water ? **1**

(A) 0.1 M Glucose
(B) 0.1 M $CaCl_2$
(C) 0.1 M KC*l*
(D) 0.1 M Urea

3. Which of the following is <u>not</u> a transition metal ? **1**

(A) Sc
(B) Ag
(C) Hg
(D) Cu

4. Which of the following represents the fraction of molecules with energies equal to or greater than $E_a$ ? **1**

(A) $+\frac{E_a}{RT}$
(B) $e^{-E_a/RT}$
(C) $-\frac{E_a}{RT}$
(D) $e^{+E_a/RT}$

5. What will happen during the electrolysis of aqueous solution of $CuCl_2$ by using platinum electrodes ? **1**

(A) Cu will deposit at Anode
(B) $H_2$ gas will be released at cathode
(C) $O_2$ gas will be released at anode
(D) $Cl_2$ gas will be released at anode

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For questions number **13** to **16**, two statements are given – one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below :

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

13. **Assertion (A) :** It is not possible to separate the components of an azeotrope by fractional distillation. **1**

**Reason (R) :** Components of an azeotrope have the same composition in liquid and vapour phase and boil at a constant temperature.

14. **Assertion (A) :** The presence of –OH group in phenols directs the incoming group to meta position in the ring. **1**

**Reason (R) :** –OH group in phenols activates the aromatic ring towards electrophilic substitution reaction.

15. **Assertion (A) :** Actinoids show wide range of oxidation states. **1**

**Reason (R) :** Actinoids are radioactive in nature.

16. **Assertion (A) :** The pentaacetate of glucose does not react with $H_2N - OH$. **1**

**Reason (R) :** It indicates the presence of free – CHO group in glucose.

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## SECTION – B

17. What type of deviation from Raoult's law is shown by a mixture of phenol and aniline ? Give reason.

What will happen to the boiling point of the solution on mixing phenol and aniline ? **2**

18. Why are haloarenes less reactive towards nucleophilic substitution reaction ? Give two reasons. **2**

19. (a) Write IUPAC names of the following coordination compounds : **2 × 1**

(i) $[Ag(NH_3)_2]\ [Ag(CN)_2]$

(ii) $K_3[Fe(C_2O_4)_3]$

**OR**

(b) (i) Give a chemical test to show that $[Co(NH_3)_5SO_4]Cl$ and $[Co(NH_3)_5Cl]SO_4$ are ionisation isomers. **2 × 1**

(ii) What is meant by the 'Chelate effect' ? Give an example.

20. Differentiate between the following : **2 × 1**

(i) Peptide linkage and Glycosidic linkage

(ii) Essential amino acids and Non-essential amino acids

21. Following reaction takes place in one step : **2**

$2A + B \longrightarrow 2C$

How will the rate of above reaction change if the volume of the reaction vessel is decreased to one third of its original volume ? Will there be any change in the order of reaction with the reduced volume ?

{}

## SECTION – C

22. For the first order thermal decomposition reaction, following data was obtained : **3**

$C_2H_5Cl(g) \longrightarrow C_2H_4(g) + HCl(g)$

| S. No. | Time(s) | Total Pressure (atm) |
|---|---|---|
| 1 | 0 | 0.30 |
| 2 | 30 | 0.50 |

Calculate rate constant. [Given : log 3 = 0.48]

23. (a) Answer the following : **3 × 1**

(i) Why is the Equilibrium Constant ($K_c$) related to $E^\circ_{cell}$ and not to $E_{cell}$ ?

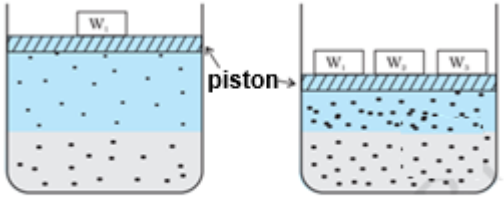
(ii) Two metals 'A' and 'B' have standard electrode potential values of −0.24 V and +0.80 V respectively. Which of these will liberate hydrogen gas from dil. $H_2SO_4$ ?

(iii) Write the cell reaction which occurs in lead storage battery when it is in charging.

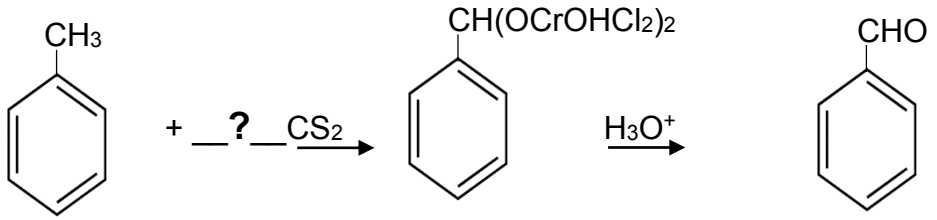
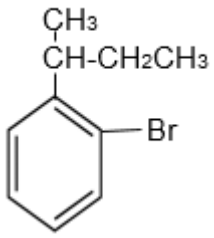
**OR**

(b) What type of battery is Mercury cell ? Why it is more advantageous than dry cell ? Write overall reaction taking place in Mercury cell. **3**

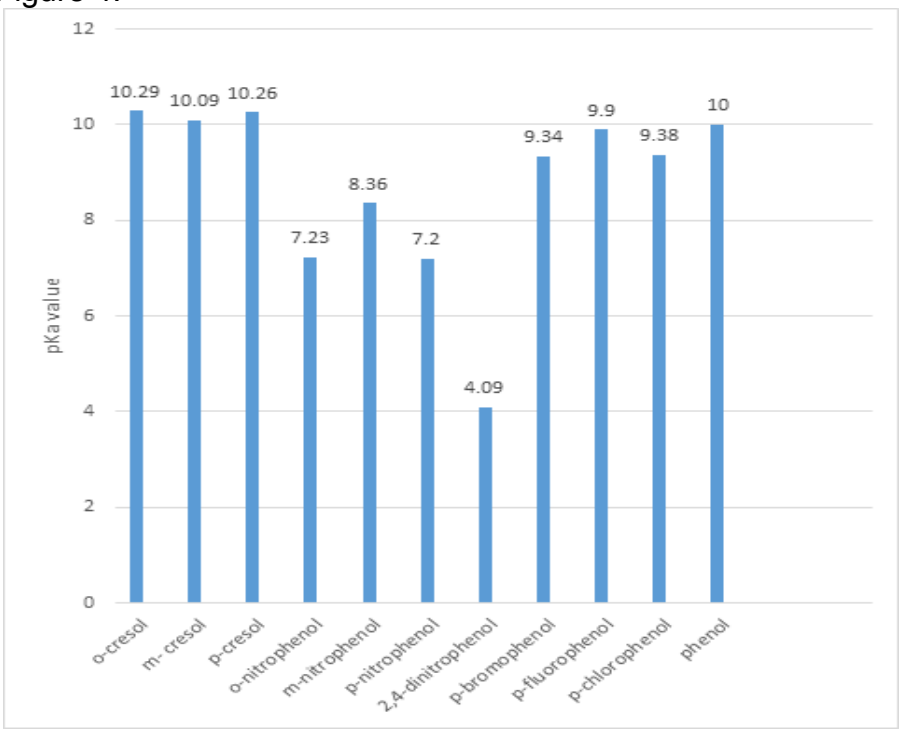
24. Calculate the boiling point of a solution containing 0.61 g of benzoic acid (Molar mass = 122 g $mol^{-1}$) in 5 g of $CS_2$ in which it dimerises to the extent of 88%. The boiling point and $K_b$ of $CS_2$ are 46.2 °C and 2.3 K kg $mol^{-1}$ respectively. **3**

| | | |
|---|---|---|
| 3 | The quantity of charge required to obtain one mole of metal "M" from $M_2O_3$ is:<br>A. 1F<br>B. 6F<br>C. 3F<br>D. 2F | 1 |
| 4 | An organic compound "X" on reaction with chloroform and alcoholic KOH gives foul smelling product.<br><br>In which of the following case compound X would be produced:<br>A. Chloromethane is treated with AgCN.<br>B. Nitrobenzene is reduced.<br>C. Benzene diazonium chloride is treated with Cu/KCN<br>D. Aniline reacts with Acetyl chloride. | 1 |
| 5 | Consider a system in a state of dynamic equilibrium, as shown in Fig. 1(a). The lower part is solution and the upper part is gaseous system at pressure p and temperature T. If the pressure is increased over the solution phase by compressing the gas to a smaller volume as shown in Fig 1(b).<br>  <br><br>Figure 1<br><br>Which of the following statements is true about figure 1(b)<br>A. solubility of gas will increase until dynamic equilibrium is reached<br>B. solubility of gas will decrease until dynamic equilibrium is reached<br>C. solubility of gas will increase until static equilibrium is reached<br>D. solubility of gas will decrease until static equilibrium is reached<br><br>***(For Visually Challenged Learners)***<br>According to Henry's law partial pressure P of a gas above the liquid at constant temperature is directly proportional to<br>A. volume of the gas<br>B. mole fraction of the gas in the solution<br>C. atmospheric pressure<br>D. vapour pressure of the solution | 1 |
| 6 | Aavya carries out hydrolysis of acetyl chloride. The reaction is represented by the following equation:<br><br>$CH_3COCl + H_2O \longrightarrow CH_3COOH + HCl$<br><br>She carries out the reaction with (i) equimolar amounts of acetyl chloride and water (ii) excess of water. The rate law obeyed in the two cases will be:<br>A. (i) First order (ii) Second Order<br>B. (i) First order (ii) First order<br>C. (i) Second order (ii) Second order<br>D. (i) Second order (ii) First order | 1 |

* There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.

| | | |
|---|---|---|
| | (II)   <br> (III) >C=O $\xrightarrow[-H_2O]{NH_2NH_2}$ ?_____ $\xrightarrow[Heat]{KOH/\text{ethylene glycol}}$ >CH2 + N2 | |
| 25 | (I) Aman prepared a solution by dissolving 10 ml of Acetone and 10 ml of Carbon disulphide. The net volume of resulting solution was not found to be 20 mL.<br>a) What would be the observation in this case? Comment.<br>b) In case carbon disulphide is replaced by chloroform in the above case, what would be the expected changes in volume of the resulting solution? Support your answer with appropriate reason. | 1+1 |
| | (II) Calculate the concentration of oxygen (in ppm) present in 1 litre of sea water. Given that a litre of sea water weighs 1030 g and contains about $6 \times 10^{-3}$ g of dissolved oxygen. | 1 |
| 26 | (I) Convert the following;<br>a) Nitrobenzene to flourobenzene<br>b) Benzoic acid to benzylamine | 2 |
| | (II) Aniline undergoes electrophilic substitution reactions readily. However, acetylation of aniline is required to obtain monosubstituted products. Give reason for your answer | 1 |
| | **Attempt either A or B** | |
| 27(A) | (I) Write the IUPAC name of the following compound   | 1 |
| | (II) When $CH_3CH_2CH_2Cl$ reacts with AgCN and KCN, Compound "X" and "Y" respectively are formed. Identify "X" and "Y". Support your answer with appropriate reason. | 2 |
| | **OR** | |
| 27(B) | (I) What is the IUPAC nomenclature of the compound: $(CH_3)_3C\text{-}C(CH_3)Br\text{-}CH_2CH_3$ | 1 |

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| | | |
|---|---|---|
| | (II) Which of the following will be hydrolysed more readily?<br>$CH_2=CH-CH_2Cl$ or $(CH_3)_2CHCl$<br>Support your answer with reason. | 2 |
| 28 | (I) On adding 100 g of non-volatile solute A in 1000 g of water, the vapour gets reduced by 50 percent. Calculate the molar mass of solute A.<br><br>(II) It has been observed that nitric acid solution containing 32% of water by mass, cannot be separated into its components through fractional distillation. Why? | 2<br><br>1 |
| **Section D**<br>Question No. 29 & 30 are case-based/data -based questions carrying 4 marks each. | | |
| 29 | **Acidity of substituted phenols**<br>Phenol and substituted phenols are relatively more acidic than aliphatic alcohols. The pKa value is a quantitative measure of the acidic strength. The effect of substituents on acidity of phenols (pKa values) is given in Figure 4.<br>  <br><br>**Figure 4**<br>**Study the graph and answer the following questions**<br><br>(I) The pKa value of 2,4,6- trinitrophenol will be:<br>A. more than 8.36<br>B. between 7.2 and 8.3<br>C. between 4.09 and 2.3<br>D. less than 4.09<br><br>(II) The order of acidity of p-halophenols is :<br>A. p-bromophenol > p-chlorophenol > p-flourophenol > phenol<br>B. phenol > p-flourophenol > p-chlorophenol > p-bromophenol<br>C. p-chlorophenol > p-bromophenol > p-flourophenol > phenol | <br><br><br>